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RESEARCH ARTICLE

Spatiotemporal population dynamics of the Caddo Madtom (*Noturus taylori*), a narrow-range endemic of the Ouachita Highlands

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Abstract

The Caddo Madtom, *Noturus taylori*, is endemic to three river drainages of the Ouachita Highlands in the southeastern United States. Conservation concern for *N. taylori* has been heightened by recent studies based on analyses of allozyme data suggesting population decline and a possible extirpation event, which lead to *N. taylori* petitioned for listing under the Endangered Species Act in 2011. The objective of this study was to determine if contemporary factors, historic processes, or a combination of both, using microsatellite DNA loci and mitochondrial (mt) DNA, have influenced patterns of genetic structuring for *N. taylori* to better inform conservation management strategies. Phylogeographic and Bayesian cluster analyses suggest that genetic structuring between Caddo and Ouachita river drainages was shaped by historical processes prior to construction of reservoir dams that dissect the drainages. We found no evidence that reservoir dams influenced contemporary patterns of genetic structure, thus it is likely that larger river courses, and life-history characteristics conducive to headwater habitats, play important roles in limiting gene flow between drainages, and to some extent within drainages. Genetic diversity of *N. taylori* was relatively high compared to other endangered madtom species; however, compared to the Ouachita River population, genetic diversity was significantly lower for the Caddo River population. Collectively, these results suggest that *N. taylori* populations are not heavily impaired, and the genetic variation and structuring is most attributable to historic processes. However, the endemic status and narrowly fragmented distribution still renders *N. taylori* populations vulnerable to extirpation or extinction by stochastic events.

Keywords Conservation · Habitat fragmentation · Microsatellites · Phylogeography

Introduction

For mainland aquatic endemic species, genetic structuring is often analogous to island species with drainage-delimited distributions because of limited dispersal and migration abilities, which decreases genetic variation (Frankham 1997). Pleistocene climate fluctuation is hypothesized to have facilitated rapid genetic change in North American freshwater fish taxa (Ray et al. 2006; Near et al. 2011); however, the genetic structuring of freshwater fish taxa within lowland habitat of the Gulf Coastal Plain were not as influenced by Pleistocene climatic changes, resulting in fewer examples of genetically structured populations throughout the Mississippi River basin when compared to species inhabiting highland geographic regions (Avise et al. 1998; Elderkin et al. 2008). Though historic geologic events have a major influence on the genetic composition within and among North American freshwater organisms, intrinsic factors (e.g., dispersal ability and habitat preference) and extrinsic barriers (e.g., dams and larger order rivers) can also affect genetic structuring (Zickovich and Bohonak 2007; Wilson et al. 2004; Fluker et al. 2014). A major component of biodiversity conservation includes preservation of genetic diversity within taxa (IUCN 2016); therefore, it is necessary to understand the implications of limited genetic variation within and among populations of endemic species since lower genetic diversity may inhibit species' ability to adapt to environmental change (Frankham 1997).

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The Caddo Madtom, *Noturus taylori* (Siluriformes: Ictaluridae), is a small, benthic catfish species endemic to the headwaters of the Caddo, Little Missouri, and Ouachita river drainages within the Ouachita Mountains Ecoregion, USA (Fig. 1a). *Noturus taylori* is a Species of Greatest Conservation Need in Arkansas (SGCN; Fowler and Anderson 2015) and was among the 404 southeastern U.S. species petitioned by the Center for Biological Diversity for federal listing under the Endangered Species Act (ESA; USFWS 2011). The basis for state and possible federal protection for *N. taylori* was from a previous conservation assessment (Turner and Robison 2006) and listing recommendations from the International Union for the Conservation of Nature (IUCN) Red List, NatureServe, and the American Fisheries Society (AFS). NatureServe and IUCN are quantitative conservation status listing formats that are widely used throughout the conservation community, and AFS reports the status of North American imperiled freshwater biota that is comparable in conservation ranking (see Jelks et al. 2008). The conservation statuses for *N. taylori* are endangered (IUCN 2016), threatened (AFS, Jelks et al. 2008), and G1 (NatureServe 2019), which is a global rank defined as

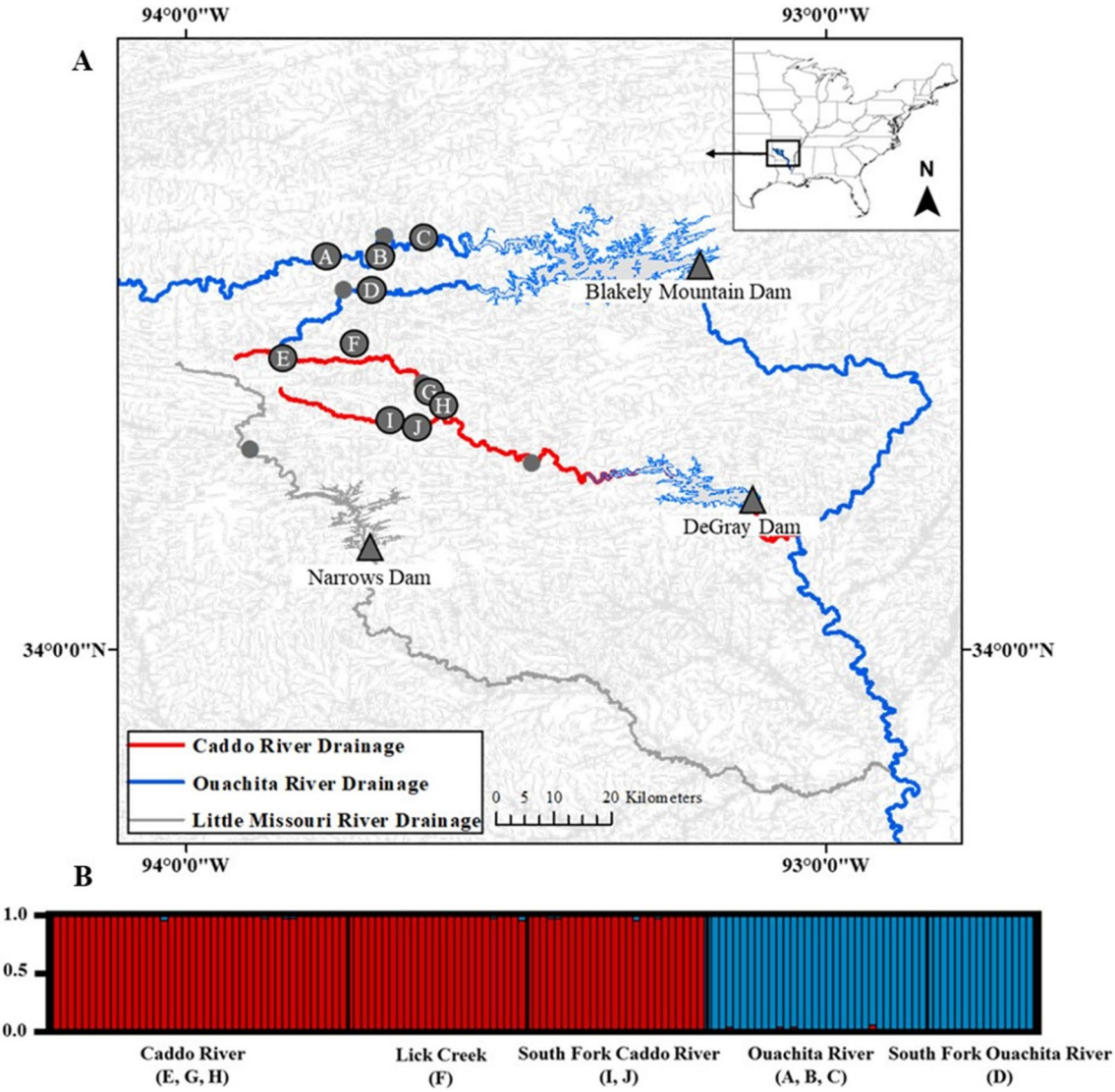

Fig. 1 **a** Species distribution and sampling localities for *Noturus taylori*. All symbols represent sampled historic localities. Lettered circles represent detection of *N. taylori* and collected tissue samples. Ouachita River samples: [A] Ouachita River at Oden Access; [B] Ouachita River at Rocky Shoals Recreation Area; [C] Ouachita River at River Bluff Recreation Area; and [D] South Fork Ouachita River sample. Caddo River samples: [E] headwaters of the Caddo River; [F] Lick Creek; [G] Collier Creek; [H] Caddo Gap; and [I and J] South Fork Caddo River mainstem. **b** Genetic structuring of *Noturus taylori* populations. STRUCTURE analysis bar plot inferred from microsatellite DNA data depicting the best-fit genetic structure [$K=2$]. Bars represent the multilocus genotypes of individuals, and colors correspond to the probability of ancestry to each cluster

critically imperiled. The conservation concern is attributed to habitat destruction and modification in combination with a restricted distribution (Turner and Robison 2006; Jelks et al. 2008; USFWS 2011).

The last record of *N. taylori* from the Little Missouri River drainage was from 1978 (Robison and Harris 1978), where the species is now thought to be extirpated (J.L. Harris, pers. comm.). A previous allozyme-based study of *N. taylori* revealed that large-reservoir dams were negatively impacting genetic connectivity for *N. taylori* populations in the Caddo and Ouachita rivers (Turner and Robison 2006). Additionally, Turner and Robison (2006) reported low allelic diversity, which is indicative of small effective population sizes, as well as significant genetic structuring and cessation of gene flow between the Caddo and Ouachita river drainages. Turner and Robison (2006) estimated that divergence among Ouachita River populations paralleled the closure of Blakely Mountain Dam (1953; USACE 2016), which impounds the Ouachita River. The genetic differentiation observed throughout the Ouachita River drainage occurred between populations from the mainstem of the Ouachita River and the South Fork Ouachita River. However, divergence observed among Ouachita and Caddo River populations predated the closure of DeGray Dam (1969; USACE 2016), which impounds the Caddo River (Turner and Robison 2006).

The discovery and description of *N. taylori* followed the closures of both Blakely Mountain and DeGray dams (Douglas 1972), and there is limited distribution data for the species prior to the construction of these dams. The Caddo and Little Missouri rivers are larger tributaries to the Ouachita River; therefore, it is possible that the species was once widespread throughout the basin and became isolated with the consecutive closures of reservoir dams. In addition to Blakely Mountain and DeGray dams, Narrows Dam impounds the Little Missouri River (constructed in 1950; USACE 2016), and there are two additional reservoir dams downstream of Blakely Mountain Dam on the Ouachita River, Carpenter and Remmel dams (constructed in 1932 and 1920, respectively; USACE 2016). Therefore, isolation of *N. taylori* populations may have occurred over a 90-year period, further influencing genetic structuring, decreasing genetic variation, and limiting dispersal ability among populations.

Data from Turner and Robison (2006) provided the first evidence that *N. taylori* does not share recent gene flow between major drainages (i.e., between Ouachita and Caddo rivers); however, small sample sizes precluded confident pattern-process interpretations of genetic structure and variation over finer spatiotemporal scales. Thus, a more extensive conservation genetic assessment is needed to evaluate whether the observed population genetics are a result of dams or other contemporary anthropogenic processes, or historically natural processes in order to provide more information for long-term conservation planning. In this study, two primary methods were used to estimate the spatiotemporal population dynamics of *N. taylori*. (1) We used phylogeographic analyses of mitochondrial DNA (mtDNA) to reconstruct the evolutionary history of *N. taylori* populations. Coalescent-based models were used to reconstruct demographic changes through time to obtain a better understanding of the observed genetic variation among *N. taylori* populations. Since there is evidence that gene flow has ceased among drainages, in addition to support for historic processes affecting genetic structure between Caddo and Ouachita River populations (Turner and Robison 2006), we predict that populations in the respective drainages are reciprocally monophyletic. (2) Analyses of nuclear microsatellite DNA loci were used to elicit further understanding of genetic structure and patterns of gene flow among *N. taylori* populations. With the lack of collection data prior to the reservoir closures and evidence of the reservoirs influencing *N. taylori* population dynamics (Robison and Harris 1978; Turner and Robison 2006), we predict that contemporary estimates of *N. taylori* population genetic structure will reflect isolation patterns influenced by the closure of the reservoir dams.

Materials and methods

Sample collections

Surveys were conducted at all accessible historic localities ($N = 15$; Fig. 1a) derived from the Fishnet2 database portal (www.fishnet2.net) and previous collection surveys (Harris and Douglas 1978; Robison and Harris 1978; Hardman 2004; Turner and Robison 2006). Individuals were successfully collected by seine and electrofishing from four localities in the Ouachita River and six localities in the Caddo River (Table 1). Tissue samples were collected from ≥ 30 individuals per locality as fin clips and stored in vials of 95% EtOH for further processing. Most individuals were released

Table 1 Samples used in genetic analyses of *Noturus taylori*

Populations by drainage/locality	Map ID	*N*mt	*N*m
Ouachita River drainage			
Ouachita River	A, B, C	10	31
South Fork Ouachita River	D	7	15
Caddo River drainage			
Caddo River	E, G, H	20	41
Lick Creek	F	10	25
South Fork Caddo River	I, J	9	25

Map ID corresponds to Fig. 1. Number of individuals used in the mitochondrial DNA analyses (*N*mt) and microsatellite analyses (*N*m)

live after fin clipping, but select individuals were preserved in 10% formalin as vouchers and deposited in the Arkansas State University Museum of Zoology (ASUMZ).

Microsatellite genotyping and DNA sequencing

Genomic DNA was extracted following the Qiagen DNEasy kit protocol (Qiagen Inc., Germantown, MD). A total of 137 individuals were genotyped for 11 microsatellite DNA loci, with primers designed for the Yellowfin Madtom (*Noturus flavipinnis*; NflA9, NflC122, NflD109, NflD123, and NflD140; Williams and Moyer (2012); Online Resource 1) and the Slender Madtom (*Noturus exilis*; Noex210, Noex216, Noex217, Noex222, Noex223, and Noex233; Online Resource 1). Polymerase chain reaction (PCR) amplifications were conducted in 12.5-µl reaction volumes, using either 5′-6-FAM or 5′-HEX fluorescently labeled forward primers and unlabeled reverse primers and followed protocols of Fluker et al. (2010). However, for primer sets Noex217, Noex223, and Noex233, a three-primer method was used, with one unlabeled primer, one primer with a 5′ tag (CAG sequence: 5′-CAGTCGGGGTCATCA-3′), and a third primer consisting only of the CAG tag with either the 5′-6-FAM or 5′-HEX fluorescent label (Online Resource 1), following protocols of Boutin-Ganache et al. (2001) and Hudman et al. (2008). Fluorescently labeled fragments were mixed with 0.5-µl GeneScan™ 500 ROX™ dye size standard (Applied Biosystems), 9.0-µl Hi-Di Formamide, and read on an Applied Biosystems model 3730xL DNA Analyzer. Peak Scanner™ Software v1.0 (Applied Biosystems) was used to obtain DNA fragment size, and prior to statistical analyses, alleles were binned with the program FLEXIBIN 2.0 (Amos et al. 2007) and screened for scoring errors and null alleles with MICRO-CHECKER (Van Oosterhout et al. 2004).

Approximately 10 individuals per sub-drainage basin (i.e., Ouachita, South Fork Ouachita, Caddo, and South Fork Caddo rivers, and a tributary to the Caddo River; Table 1) were amplified for the mitochondrial cytochrome *b* gene (cyt *b*) using primers of Song et al. (1998) and PCR amplification protocols of Lang and Mayden (2007). PCR products were purified using the QIAquick PCR purification kit (Qiagen) and purified products were sequenced on an Applied Biosystems 3730xL DNA Analyzer. Alignments were completed manually using BioEdit v7.2.5 (Hall 2011).

Phylogeographic analyses of drainage scale structure

The best model of nucleotide substitution (GTR+I) for *N. taylori* cyt *b* sequences was determined with jModelTest (Darriba et al. 2012), using the Akaike Information Criterion. Bayesian inference was used to generate a phylogeographic hypothesis of the drainage scale structure among *N. taylori* populations using MrBayes 3.2.7 (Huelsenbeck and Ronquist 2001). To root the tree, outgroup taxa were *Cranoglanis bouderius*, *Ameiurus catus*, *A. serracanthus*, *Ictalurus punctatus*, *I. furcatus*, *N. elegans*, *N. elutherus*, *N. furiosus*, *N. munitus*, and *N. placidus*. All GenBank accession numbers for outgroup taxa and *N. taylori* sequences generated in this study can be found in the supplemental material (Online Resources 2). Analyses were run with 1.00×10^6 generations and sampled trees every 100 generations. Two independent runs, each with four simultaneous chains were completed, and the first 25% of generations were discarded as burn-in. Analyses continued until average standard deviation of split frequencies were below 0.01, and the potential scale reduction factor was close to 1.00 for all parameters. Parameter files were further assessed in Tracer (Rambaut et al. 2007) for convergence. Significance was determined at $\alpha = 0.05$. A haplotype network was constructed using TCS v1.2 (Clement et al. 2000). Haplotype networks were constructed with the statistical parsimony algorithm, which estimates the number of differences of single nucleotide substitutions among haplotypes. Genealogical relationships among *N. taylori* populations were assessed using all 56 cyt *b* sequences.

Genetic diversity and population structure analyses

Deviations from Hardy–Weinberg equilibrium (HWE) and the possibility of linkage disequilibrium (LD) for microsatellite DNA loci were assessed with GENEPOP v4.2 (Raymond and Rousset 1995). Probabilities of significance for departure from HWE and LD predictions were estimated with Markov chain parameters using 5000 dememorizations, 200 batches, and 5000 iterations per batch. Genetic diversity was estimated in Arlequin v3.5 (Excoffier and Lischer 2010) and measured as the mean number of alleles per locus (A), observed heterozygosity (H_o), and expected heterozygosity (H_e). Since sample size was small for one population, allelic richness (AR) was estimated as number of alleles per locus independent of sample size with HP-RARE (Kalinowski 2005), in addition to the number of private alleles per drainage (A_p). Estimates of genetic diversity were assessed for mean differences within and among drainages using a Welch's t-test, as well as a nonparametric Wilcoxon Mann–Whitney test in RStudio v1.0.136 (RStudio Inc., Boston, MA).

Genetic structure was assessed independently for microsatellite and mtDNA data using an analysis of molecular variance (AMOVA; Excoffier et al. 1992) in Arlequin. One analysis was conducted with all localities in one group irrespective of drainage, and another was conducted with one hierarchal category with two groups of localities associated by drainage (i.e., Caddo River group and Ouachita River

group). The AMOVAs, as well as tests for significant F_{st} values, were assessed with 10,000 permutations. Correlation between genetic and geographic distance was assessed with F-statistics using the Mantel test in Arlequin for both microsatellite DNA loci and mtDNA datasets. Distance was measured in river kilometers between sampled localities, using ArcGIS v. 10.3.1 (ESRI 2015). The Mantel test was assessed with 10,000 permutations.

A Bayesian cluster analysis was conducted with microsatellite data in STRUCTURE v2.3 (Pritchard et al. 2000) to determine the number of genetically distinct clusters (*K*). Clusters were assigned based on allele frequencies without a priori population information and an admixture model that allows admixture of genotypes among populations (Pritchard et al. 2000). Starting *K* values ranged from one to six, and ten runs were performed with 5.00×10^5 iterations and a burn-in of 5.00×10^4 for each cluster estimate. Evaluations of *K* were determined using both the Evanno ad hoc approximation (ΔK; Evanno et al. 2005) and by examination of the mean estimated *Ln* probability of the data (*LnPD*) with the program STRUCTURE HARVESTER (Earl and vonHoldt 2012). The Evanno method cannot assess $K = 1$; therefore, an additional clustering analysis using a cross-entropy criterion, which is also used by the program ADMIXTURE (Alexander and Lange 2011), was performed in RStudio with the package LEA (Frichot and François 2015).

Contemporary and historic gene flow analyses

Migration among populations was estimated with microsatellite DNA loci using BayesAss v3.0 (Wilson and Rannala 2003) and Migrate-n v3.2 (Beerli 2006). BayesAss implements a Bayesian multilocus method that estimates contemporary patterns of immigration rate per generation (*m*) among populations, and HWE is not assumed. BayesAss assumes both linkage disequilibrium and that migration and genetic drift do not change subpopulation allele frequencies over the previous 2–3 generations (Wilson and Rannala 2003). Analyses followed recommendations of Faubet et al. (2007), with 10 runs of 10^7 iterations, a 10^6 burn-in, and a random seed for each run. To ensure optimal parameter space was searched, delta values were adjusted accordingly (e.g., within 40–60% acceptance rate of iterations; Wilson and Rannala 2003). Convergence of parameter estimates in the best runs was assessed by manual examination of trace files in Tracer.

Migrate-n uses coalescent-based methods to estimate historic patterns of pairwise gene flow ($M = m/\mu$), and the program assumes an island model with equilibrium parameters (Beerli and Felsenstein 2001), allowing for estimation of unequal population size and nonsymmetrical gene flow. These estimates are assumed constant; therefore, estimates of migration are best considered as indicators of long-term patterns. Analyses were performed with a maximum-likelihood approach that applied the Metropolis-coupled Markov chain Monte Carlo search parameters (Beerli 2009). Preliminary runs were conducted with F_{st} to determine starting parameters for subsequent runs. Statistical significance of demographic parameters was estimated from five additional runs (referred to as final runs hereafter). Search parameters for the final runs included 10 short chains of 50,000 generations, 2000 recorded genealogies, and a burn-in of 10^6 generations, as well as three long chains of 5.00×10^6 generations with 5000 recorded genealogies. All chains used the Brownian motion approximation as the mutation model, as well as static heating to maximize the search space (e.g., 1.00, 1.50, 3.00, and 10^5). Convergence was assumed accurate when the 95% confidence intervals were overlapping in at least three of the final runs (Beerli 2009).

To compare migration rates, *M* was converted to proportion of immigrants (*m*) for populations with the previous formula, where mutation rate (μ) is equal to 5.50×10^{-4}. The mutation rate was derived from estimates of microsatellite mutation rates for the Common Carp (*Cyprinus carpio*; Yue et al. 2007). Currently, there are no estimates of microsatellite mutation rates for madtom catfishes, or catfishes in general; however, the estimated rate derived from Yue et al. (2007) is within the range of microsatellite mutation rates reported for vertebrates by Ellegren (2000) and Xu et al. (2000) (i.e., 0.001–0.0006). Historic and contemporary estimates were evaluated with a Wilcoxon signed-rank test, in addition to differences in historic and contemporary patterns of between-drainage migration.

Heterozygosity-excess tests were conducted with microsatellite DNA loci in BOTTLENECK v1.2.02 (Piry et al. 1999) to determine whether populations have experienced recent decline in effective population size. A two-phase model (TPM) was used with 95% single-step mutations and 12% variance of multi-step mutations (Piry et al. 1999). A sign test and Wilcoxon signed-rank test were implemented within the program to test if populations exhibited a significant number of loci with heterozygosity excess. However, these methods have low power, especially with small numbers of loci (i.e., less than 20) unless the decline is > 90% or has occurred recently (i.e., within the last few generations; Williamson-Natesan 2005).

Results

Phylogeography and drainage scale structure

The MrBayes analysis using mtDNA failed to recover reciprocally monophyletic clades at the drainage scale (Fig. 2a). However, an important feature of the phylogram is the geographical associations of haplotypes. All

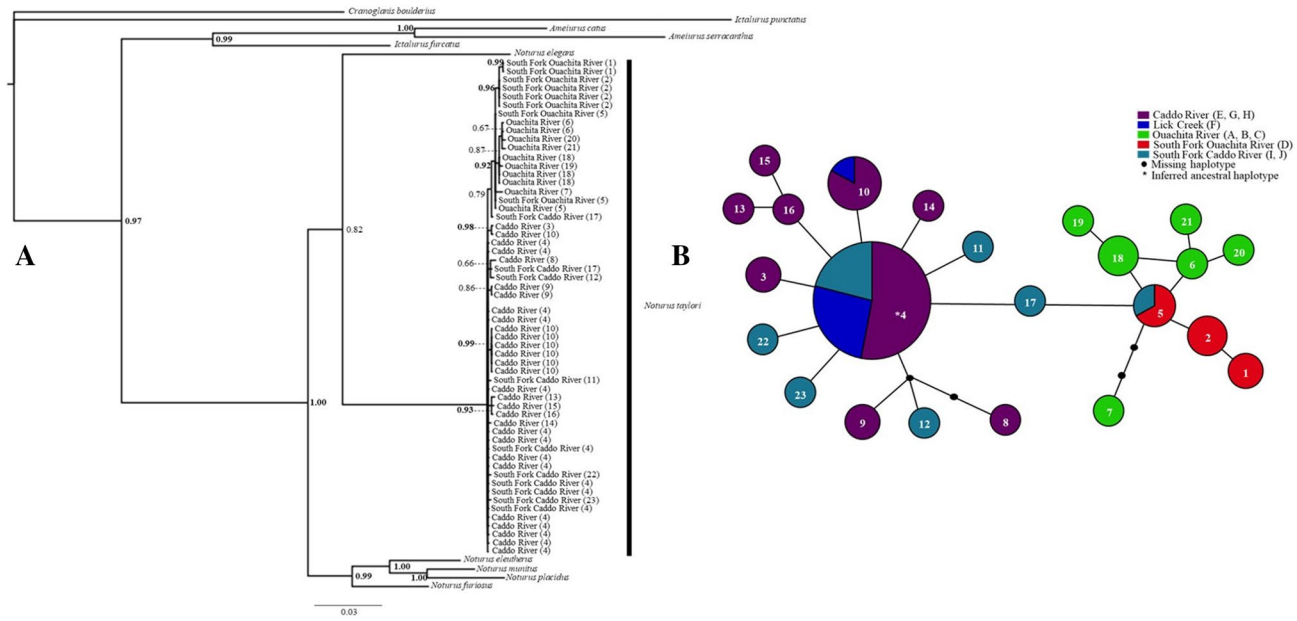

Fig. 2 **a** MrBayes inferred phylogram for 56 *Noturus taylori* mitochondrial cytochrome *b* sequences. The posterior probabilities are presented at each node. Bolded posterior probabilities imply significance at α = 0.05. Numbers in parentheses correspond to haplotype numbers in the haplotype network. The scale bar represents substitutions per site. **b** Mitochondrial DNA haplotype network generated by statistical parsimony analysis in TCS. Circles (haplotypes) are proportional to the sampled individuals with the respective haplotypes, lines represent one mutational step between haplotypes, and black circles on lines represent an additional mutational step

haplotypes from the Ouachita and South Fork Ouachita rivers were resolved in a monophyletic clade, albeit with low posterior probability support of 0.92. The remaining clades with significant support consisted of haplotypes sampled from the same, or geographically proximate, localities.

Phylogeographic reconstruction of sub-drainage basin structure

Twenty-three mtDNA haplotypes were identified among the 56 individual *N. taylori* sequenced for cyt *b*. The TCS analysis resulted in a haplotype network with two haplogroups, each with strong correspondence to either Caddo or Ouachita river drainages (Fig. 2b; Online Resource 3). Castelloe and Templeton (1994) suggested that haplotypes with numerous mutational connections and the highest frequency within a sample would have the highest root probability, inferring a retained ancestral haplotype. Therefore, haplotype 4 in the Caddo River population and haplotype 5 in the Ouachita River population are speculated to be retained ancestral haplotypes within the two populations. Except for one haplotype shared between the drainages (haplotype 5), each drainage has unique sets of haplotypes, and the divergence of all haplotypes are relatively recent, with few mutational differences between haplotypes (Fig. 2b).

Genetic diversity and population structure

Ten of the eleven microsatellite loci were used for analyses. MICRO-CHECKER indicated a presence of null alleles for multiple populations with the marker Noex233, so it was removed from subsequent analyses. With 50 locus-population combinations, 18 significantly (α = 0.05) deviated from HWE due to a deficiency of heterozygotes; however, following Bonferroni correction (α = 0.001), six locus-population comparisons deviated from HWE. There was no distinct pattern of deviation within a population or with a locus, and no evidence of linkage disequilibrium. Therefore, no other locus was removed from subsequent analyses. When assessing differences in genetic diversity (A, AR, H_o, H_e) within and among drainages, no significant differences were found at an α = 0.05 (Online Resources 4), with the exception of between drainage comparison of H_e ($W = 155.0$; $p = 0.01$), with the Caddo River drainage having a lower H_e than the Ouachita River drainage. Genetic diversity was similar among all other *N. taylori* populations (Table 2).

The STRUCTURE analysis found the most likely number of clusters to be $K = 2$ ($\Delta K = 4295.26$; Online Resources 6). Results of the STRUCTURE analysis corroborate the results of the structuring analysis in the R-program, LEA, invalidating the plausibility of $K = 1$ for this dataset (Online Resources 5). The $K = 2$ configuration from STRUCTURE runs supported a high degree of population structure

Table 2 Genetic diversity estimates (mean for 10 microsatellite loci) for *Noturus taylori* with comparisons to *N. flavipinnis* and *N. baileyi*

Species/populations	N	A	AR	A_p	H_o	H_e
Noturus taylori						
Ouachita River	31	8.50	7.11	1.29	0.67	0.74
S. Fork Ouachita River	15	7.75	6.28	0.94	0.69	0.77
Ouachita River drainage mean	–	**8.13**	**6.70**	**1.12**	**0.68**	**0.76**
Caddo River	41	7.70	5.99	0.62	0.62	0.65
Lick Creek	25	5.40	4.75	0.46	0.57	0.55
S. Fork Caddo River	25	6.00	5.15	0.40	0.66	0.63
Caddo River drainage mean	–	**6.37**	**5.30**	**0.49**	**0.62**	**0.61**
Noturus flavipinnis						
Tennessee River drainage mean	56	**1.51**	–	–	**0.11**	**0.11**
Noturus baileyi						
Tennessee River drainage mean	87	**2.33**	–	–	**0.17**	**0.19**

Data for *N. flavipinnis* and *N. baileyi* were obtained from Williams and Moyer (2012, *unpublished*)

N number of individuals; A mean number of alleles per locus; AR allelic richness; A_p mean number of private alleles; H_o observed heterozygosity; H_e expected heterozygosity

Bold values are the mean estimates of genetic diversity for the respective drainages

between Caddo and Ouachita river drainages, with individuals assigning with high probability (> 0.95) to their representative drainage (Fig. 1b). The AMOVAs, however, supported most variation being attributable to between-population differences irrespective of drainages (F_{st} = 0.20, P < 0.001 [microsatellites]; Φ_{st} = 0.37, P < 0.001 [mtDNA]), with differentiation between drainages not being significantly different from zero (F_{ct} = 0.21, P = 0.099 [microsatellites]; Φ_{ct} = 0.42, P = 0.100 [mtDNA]). Microsatellite-based F_{st} pairwise comparisons revealed significant genetic distances between all populations, with larger distances accumulating between drainages, which is relatively consistent with the mtDNA Φ_{st} pairwise comparison (Table 3). However, the Mantel test did not support correlation between genetic and geographic distance. Additionally, though low, individuals within populations were more closely related than expected under HWE (F_{is} = 0.04, P = 0.005).

Contemporary and historic gene flow

Coalescent and Bayesian estimates of migration rates (m) were low overall (all < 1.0 immigrants per generation; Online Resources 6). There was evidence of asymmetrical contemporary gene flow (non-overlapping of confidence intervals) for two migration events; migration from South Fork Caddo River to Caddo River and from South Fork Ouachita River to Ouachita River (Online Resources 6). The Wilcoxon signed-rank test indicated that historic estimates of migration rates (Migrate-n) were significantly lower than contemporary estimates (BayesAss; W = 7, P < 0.001), but differences in historic and contemporary between-drainage migration rates were not significant (W = 1, P = 0.063). Lastly, there was no significant evidence of a recent bottleneck among *N. taylori* populations using the program BOTTLENECK.

Table 3 Estimated pairwise fixation indices (F_{st} below diagonal and Φ_{st} above diagonal) for *Noturus taylori* populations

Population	Caddo River	Lick Creek	South Fork Caddo River	Ouachita River	South Fork Ouachita River
Caddo River	–	0.000	0.018	0.468*	0.509*
Lick Creek	0.066*	–	0.000	0.434	0.667
S. Fork Caddo River	0.068*	0.141*	–	0.414*	0.475*
Ouachita River	0.227*	0.275*	0.231*	–	0.192*
S. Fork Ouachita River	0.305*	0.357*	0.312*	0.069*	–

*Indicates significance at α = 0.05

Discussion

The recent petition for listing under the ESA, as well as state-level conservation concern for *N. taylori* warranted a thorough conservation genetic assessment for the species throughout its known distributional range. The objective of this study was to estimate the spatiotemporal population dynamics of *N. taylori* using phylogeographic analyses of mtDNA to reconstruct the evolutionary history of *N. taylori* populations and analyses of nuclear microsatellite DNA loci to elicit further understanding of genetic structure and patterns of gene flow among *N. taylori* populations. We predicted that (1) populations in the respective drainages are reciprocally monophyletic, and (2) that contemporary estimates of *N. taylori* population genetic structure will reflect isolation patterns influenced by the closure of the reservoir dams. Phylogeographic analyses did not resolve *N. taylori* populations as reciprocally monophyletic at the drainage scale. However, the clustering of mtDNA haplogroups, in addition to corresponding patterns of significant pairwise F_{st} values from both mtDNA and microsatellite loci, are suggestive of isolation between Caddo and Ouachita river drainages prior to construction of reservoir dams. Low migration rate estimates further support the lack of contemporary gene flow between Caddo and Ouachita river drainages and highlights low within-drainage migration. Finally, there was evidence of significantly lower genetic diversity in the Caddo River drainage. Therefore, these findings suggest there are two unique populations of *N. taylori* that are distinct at the drainage scale. The isolation of these populations is relatively recent on an evolutionary timeframe, but preceding contemporary factors, such as reservoir closures.

Phylogeographic structure

The prediction for the phylogeographic analyses was that clades would derive from the respective river systems (i.e., Ouachita River, South Fork Ouachita River, and Caddo River clades). Briefly, Turner and Robison (2006) used allozyme markers to assess whether the closure of Blakely Mountain Dam could explain the divergence among Ouachita River drainage localities, and the results of the study found that ≤ 150 generations were needed to accumulate the observed genetic differences between the South Fork Ouachita and the mainstem Ouachita rivers. Turner and Robison (2006) set a generation equal to one year based on data for *Noturus lachneri* (Gagen et al. 1998), and generation time was determined from population genetic theory equations designed by Hedrick (2004) that approximates generation time from genetic variance accumulated among populations at a rate determined by N_e. However, the generation time required to explain the observed divergence between the Caddo and Ouachita rivers predated the closure of DeGray Dam by a few hundred years (approx. 320 generations; Turner and Robison 2006). Turner and Robison (2006) speculated that population structure of *N. taylori* most likely resulted from historic processes based on the divergence estimation of Caddo River populations. The current study supports the results of Turner and Robison (2006), with the phylogeographic reconstruction depicting some degree of historic isolation between the drainages (Fig. 2).

The phylogeny failed to recover the drainages as reciprocally monophyletic (Fig. 2a); however, clustering of all Ouachita River individuals to a single clade suggests isolation on a relatively recent timescale, with evidence of incomplete lineage sorting for *N. taylori* between drainages. Compared to the MrBayes tree, the assortment of individuals into Caddo and Ouachita river haplogroups in the haplotype network provides more support for drainage-level differences, despite one shared haplotype between drainages (haplotype 5, Fig. 2b). The haplotype arrangement corresponds to an intermediate divergence, specifically an allophyly arrangement, meaning the representative haplotypes are from two groups, generally in monophyletic subclades, though samples from the two populations are still intermixed across the tree (Omland et al. 2006). This incomplete lineage sorting is temporary and characteristic of diverging populations.

The mitochondrial marker was used to infer historical process (i.e., pre-impoundment conditions); therefore, the arrangement of haplotypes corresponding to their respective drainages is an indication of historical events likely influencing the observed genetic structuring. As inferred from the intermediate divergence patterns, divergence among the drainages is incomplete, with a South Fork Caddo River haplotype (haplotype 17; Fig. 2b) connecting the haplotype clusters, corroborating that isolation among populations is a recent event. The arrangement of unique clusters to respective drainages, however, may also indicate limited genetic connectivity among drainages. The Ouachita Highlands are susceptible to flood events (Hodge and Tasker 1995), and with the relatively close proximity between the headwater reaches of these systems, these systems can interdigitate among the low topography, allowing opportunistic individuals to move between systems in high flood events. Therefore, a historic flood event or stream capture (Robison and Harris 1978) could have allowed gene flow, resulting in the limited connectivity depicted in the haplotype network. Alternatively, haplotype 5 could be a remnant of an ancestral haplotype that has persisted by chance in the population, but more robust analyses would be required to assess this alternative.

Genetic diversity and population structure

Analyses of 10 microsatellite loci indicate that *N. taylori* has relatively high genetic diversity (A and H_e; Table 2) when

compared to two imperiled madtom species (*N. flavipinnis* and *N. baileyi*) with similar life-history characteristics and small geographic distributions (IUCN 2016). Microsatellite-based estimates of genetic diversity for *N. flavipinnis* (federally threatened) and *N. baileyi* (federally endangered) were obtained from Moyer and Williams (2012), which included five of the same loci used for *N. taylori* herein, but also included additional loci not used in this study. Since different loci were used between Moyer and Williams (2012) and the current study, and the geographic distributions for *N. flavipinnis* and *N. baileyi* are more constrained than that of *N. taylori* (Burr and Stoeckel 1999), comparisons of genetic diversity among *N. taylori*, *N. flavipinnis*, and *N. baileyi* must be taken with caution, but suggest relatively higher levels of genetic diversity for *N. taylori* (Table 2). There was a significant difference in genetic diversity between the Caddo and Ouachita River drainages, with the Caddo River exhibiting a lower level of diversity than the Ouachita River (Table 2). Such differences in genetic diversity can best be explained by differences in long-term effective population size (Frankham 1996) since we did not detect recent genetic bottlenecks.

There was evidence of substantial genetic variation throughout the geographic distribution of *N. taylori*; however, results from AMOVAs, pairwise F_{st} values, and STRUCTURE were somewhat contradictory. The AMOVA analyses with Caddo and Ouachita river individuals as the two hierarchical groupings were consistent with the allozyme-based study of Turner and Robison (2006) ($F_{ct}=0.21$ [microsatellites]; $\Phi_{ct}=0.42$ [mtDNA]; $F_{ct}=0.65$ [allozymes; Turner and Robison 2006]). Each of the data types revealed moderate levels of divergence between drainages, but none were significant. The AMOVA analyses assessing differentiation among all populations regardless of drainage revealed moderate and significant genetic differentiation, consistent with allozyme data ($F_{st}=0.20$, $P<0.001$ [microsatellites]; $\Phi_{st}=0.37$, $P<0.001$ [mtDNA]; $F_{st}=0.71$, $P<0.001$ [allozymes; Turner and Robison 2006]). Microsatellite and mtDNA-based population pairwise comparisons of F_{st} from this study are similar to allozyme data of Turner and Robison (2006), which show relatively lower differentiation within drainages and substantially higher differentiation between localities from the Caddo and Ouachita river drainages (Table 3). Genetic clusters determined in STRUCTURE ($K=2$) revealed a strong drainage effect with minimal admixture between Caddo and Ouachita river drainages, suggesting minimal gene flow has occurred on a contemporary time frame. Taken together, these patterns of population genetic differentiation are largely consistent with phylogeographical analyses, supporting distinctiveness at the drainage scale with lineage sorting at an intermediate state (i.e., allophyly; Omland et al. 2006).

Within-drainage population differences were relatively lower than among, which suggests moderate levels of gene flow among close populations; however, genetic distances were significant among populations (Table 3), but did not fit a pattern of IBD. Additionally, the relatively higher F_{st} values and lack of IBD suggest the absence of gene flow between drainages. The use of both microsatellite DNA loci and mtDNA provide a contemporary and historic assessment of gene flow, indicating that the large-reservoir dams may not have been the cause of cessation of gene flow between drainages. Instead, the migratory ability of *N. taylori*, the larger stream reaches between the drainage systems, or other life history characteristics prior to the dam closures resulted in the structure patterns observed.

Contemporary and historic gene flow

Common methods for direct comparisons of temporal gene flow among populations include comparing BayesAss and Migrate-n estimates (e.g., Chiucchi and Gibbs 2010; Ishiyama et al. 2015; Stevens et al. 2018; Winter et al. 2018). However, recent studies have demonstrated how such comparisons may not accurately reflect reality (Faubet et al. 2007; Miermans 2014; Samarasin et al. 2017). Samarasin et al. (2017) found that in scenarios with recent declines in migration, Migrate-n will underestimate historical migration rates, whereas BayesAss will overestimate these rates under similar scenarios. Therefore, the following interpretations for gene flow represent a conservative, qualitative approach. Estimates of migration are more robust when assessed with the Bayesian inference approach and is suggested by Beerli (2012). Our microsatellite data, however, were unable to reach convergence under this search strategy. Though Bayesian inference has been shown to be the preferred method, maximum likelihood has been shown to produce comparable results (Beerli 2006; Kuhner and Smith 2007).

Overall, the proportion of immigrants (m) was low (<1) (Online Resources 6). Contemporary estimates of m (BayesAss) for between-drainage migration may suggest that the reservoirs are an effective barrier to gene flow; however, the low historic estimates of m (Migrate-n) for between-drainage comparisons suggest historic processes (prior to reservoir impoundment) have also restricted gene flow. A plausible explanation could be that the larger main stem reaches have historically been a barrier for gene flow. Therefore, it can be speculated that *N. taylori* populations in the Caddo and Ouachita rivers have been historically isolated because of a combination of geographic distance, larger stream reaches impeding migration, and restricted dispersal ability of benthic dwelling species. The Wilcoxon signed-rank test revealed significant differences in m between historic and contemporary calculations ($W=7$, $P<0.001$), with increased gene flow occurring

more recently with the dams in place and asymmetric migration (represented by non-overlapping confidence intervals). These data show asymmetric migration in contemporary estimates from (1) the South Fork Caddo River to the Caddo River and from (2) the Ouachita River to the South Fork Ouachita River. Mechanistically, asymmetric gene flow is described as gene flow from source populations to recipient populations, which can indicate that gene flow is density dependent (Dias 1996). Alternatively, source populations can act as a sink in response to differences in habitat quality (Pulliam 1988; Palstra et al. 2007), with immigrants migrating from a population with high productivity to a population with low productivity and is speculated to be a mechanism to maintain persistence in those smaller populations.

Though it seems that gene flow has increased from historic estimates, *m* is too low to prevent the accumulation of adaptive alleles; therefore, local adaptations can arise and further expedite the evolutionary isolation between the populations in each drainage as long as gene flow remains minimal between the drainages (Slatkin 1975; Griswold 2006; Yeaman and Whitlock 2011).

Conservation implications

Analysis of mtDNA and microsatellite loci revealed that *N. taylori* populations in the Caddo and upper Ouachita river drainages exhibit patterns of genetic structuring shaped by historical processes prior to construction of reservoir dams that now dissect these drainages. In addition, the Caddo River population has significantly lower genetic diversity compared to the Ouachita River population. Taken together, we recommend that *N. taylori* from the Caddo and Ouachita river drainages be treated separately in future conservation and management plans. Genetic diversity of *N. taylori* populations was relatively high when compared to two other endangered *Noturus* species; however, the endemic status and narrowly fragmented distribution still renders *N. taylori* populations vulnerable to extirpation or extinction by stochastic events. Continued conservation plans for *N. taylori* would benefit from future studies that monitor the abundance and levels of genetic diversity of populations. Though this study failed to detect any recent bottleneck events, future findings that contradict the results of this study may reflect the negative consequences the reservoirs cause as an effective barrier to gene flow.

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Data availability All microsatellite DNA data and associated specimen data are available from the Dryad digital repository (https://doi.org/10.5061/dryad.kprr4xh20). All microsatellite data will be archived in the Dryad data repository upon acceptance of this manuscript. All GenBank accession numbers for outgroup taxa and *N. taylori* sequences generated in this study can be found in the supplemental material (Online Resources 2).

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